Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Identity Fraud Analytics, Predictive Modeling, Anomaly Detection, Machine Learning & Heuristics, Dimensional Data Modeling, Star/Snowflake Schemas, SCD Type 1/2, Real-Time Data Processing, Automated Data Quality Audits, API Integration, Behavioral Biometrics, Statistical Evaluation, Root Cause Analysis.

Programming Languages & Tools: Python, PySpark, SQL, Azure Cloud (Azure Data Factory, Azure Synapse, Azure SQL, Databricks), AWS (S3, Redshift), Snowflake, Scikit-Learn, XGBoost, Power BI, Tableau, Airflow, KIRO IDE.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Delivery Consultant & Identity Analytics Engineer — **May 2022 – Present**

- **Designed, developed, and deployed advanced machine learning models** utilizing XGBoost, Random Forests, and Scikit-Learn to detect synthetic identities, account takeovers, and fraudulent authentication vectors, improving system reliability by 20%.
- **Engineered scalable data pipelines and high-throughput frameworks** using Azure Data Factory, Azure Synapse, and PySpark to process high-volume real-time transactional logs and behavioral telemetry for automated identity fraud detection.
- **Implemented proactive anomaly detection systems and automated data quality firewalls** achieving 95% accuracy in mitigating ETL and operational discrepancies while reducing manual operational turnaround times by over 80%.
- **Integrated predictive analytics and advanced statistical forecasting models** such as SARIMA and custom heuristics within cloud environments to monitor model performance, false-positive rates, and optimize digital verification workflows.
- **Spearheaded Agentic AI and automated QA testing frameworks** using prompt engineering and KIRO IDE to streamline secure application programming interfaces (APIs) integration and accelerate identity verification delivery by 40%.

Graduate Research Data Scientist & Analytics Specialist — **May 2022 – Present**

- **Constructed dimensional data models and high-performance SQL schemas** implementing Star and Snowflake architectures with Slowly Changing Dimensions (SCD Type 1/2) for secure information systems management.
- **Executed complex statistical evaluation and data mining initiatives** leveraging Python and advanced machine learning algorithms to uncover patterns within large-scale datasets during graduate studies at the University of Cincinnati.
- **Developed interactive executive dashboards and analytical reporting solutions** using Power BI (DAX) and Tableau to translate massive operational datasets into actionable intelligence for stakeholders.

Engineering Professional — **May 2022 – Present**

- **Analyzed complex technical processes and operational workflows** applying rigorous engineering principles to optimize system reliability and resource utilization.
- **Collaborated within cross-functional teams to deliver technical projects** ensuring adherence to strict project timelines, meticulous documentation, and high-quality deliverables.
- **Applied structured problem-solving methodologies to troubleshoot anomalies** and streamline technical data processing pipelines in fast-paced production environments.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.